9.7.5 Wrap-Up: Reflection

1. Check all the statements that apply to your understanding.

    - ☐ I can read a line plot.
    - ☐ I need help reading a line plot.
    - ☐ I can describe and analyze data represented on a line plot.
    - ☐ I need help describing and analyzing data represented on a line plot.
    - ☐ I can determine the similarities and differences between a line plot and a histogram.
    - ☐ I need help to determine the similarities and differences between a line plot and a histogram for the same data.

Unit 9, Lesson 8: Comparing Distributions

Benchmarks

MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range.

MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.

Learning Targets

- ☐ I can find and interpret the mean and median of data sets.
- ☐ I can use mean and median to compare data sets.
- ☐ I can compare the distribution of data sets.

9.8.1 Warm-Up: Which One Doesn't Belong?

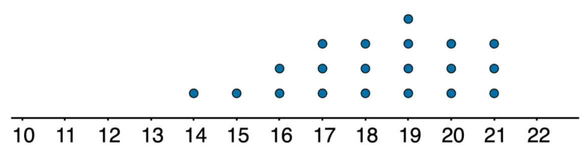
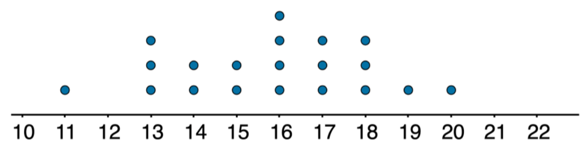
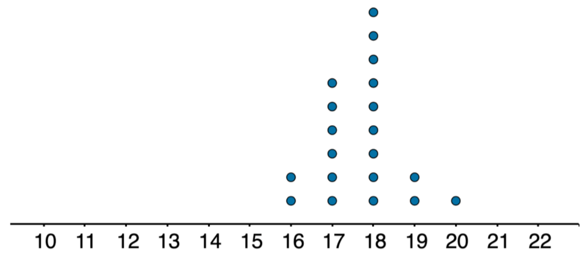
1. Which license plate doesn't belong? Explain your reasoning.

9.8.2 Collaboration: Comparing Distributions

1. Clare recorded the amount of time students in the sixth, eighth, and tenth grades spent doing homework, in hours per week. She made a line plot of the data for each grade and provided the following summary.

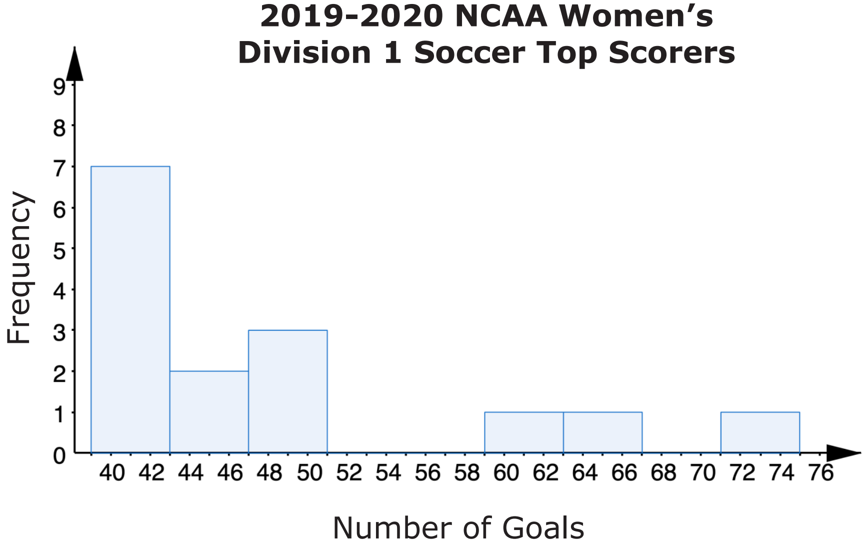
   - Students in sixth grade tend to spend less time on homework than students in eighth and tenth grades.
   - The homework times for the tenth-grade students are more alike than the homework times for the eighth-grade students.

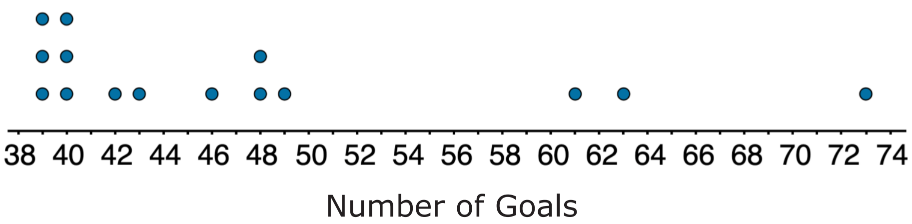
   Use Clare's summary to match each line plot to the correct grade (sixth, eighth, or tenth).

Grade Level	Line Plot
________	
________	
________	 Time (in hours)

9.8.3 Guided Practice: Comparing Distributions

1. A histogram and line plot of the number of goals scored by the top 15 women goal scorers during the 2019-2020 NCAA Women's Division 1 soccer season are shown.





a. The
- ○ histogram
- ○ line plot
- ○ histogram and line plot

can be used to determine the mean.

b. The
- ○ histogram
- ○ line plot
- ○ histogram and line plot

can be used to determine the median.

c. Which measure of central tendency do you expect to be larger, the mean or the median? Explain your reasoning.

2. The table lists the data for the top 15 women goal scorers during the 2019-2020 NCAA Women's Division 1 soccer season.

Rank	Player	Goals
1	Evelyne Viens, South Fla.	73
2	Catarina Macario, Stanford	63
3	Raimee Sherle, Boise St.	61
4	Ally Watt, Texas A&M	49
5	Tziarra King, NC State	48
5	Deyna Castellanos, Florida St.	48
7	Alexa Genas, Campbell	46
8	Morgan Weaver, Washington St.	43
9	Peyton DePriest, Middle Tenn.	42
10	Elise Flake, BYU	40
10	Samantha Dewey, Xavier	40
10	Hayley Younginer, Wake Forest	40
13	Dakota Mills, Saint Joseph's	39
13	Katie McClure, Kansas	39
13	Taylor Kornieck, Colorado	39

a. Find the mean number of goals. Interpret the mean.

b. Find the median number of goals. Interpret the median.

c. Which value is larger, the mean or the median?

d. How would the mean and median change if you found the mean goals and median goals for the top 8 scorers?

e. Find the mean number of goals for the top 8 scorers.

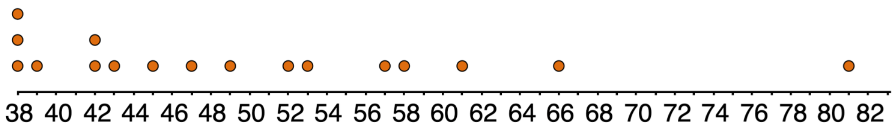
f. Find the median number of goals for the top 8 scorers.

g. Discuss with your partner how the measures of central tendency changed. Write your conclusion.

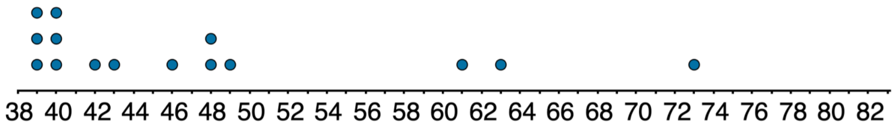
9.8.4 Your Turn!

1. The line plots show the top 15 women goal scorers during the 2016-2017 and 2019-2020 NCAA Women's Division 1 soccer season.

2016-2017 Season



2019-2020 Season



Number of Goals

a. What is different about the top 15 goal scorers during the 2016-2017 soccer season and the top 15 goal scorers during the 2019-2020 soccer season?

b. Make a hypothesis determining whether the mean of the 2019-2020 season or the mean of the 2016-2017 season is higher. Explain what influenced your hypothesis.

c. Make a hypothesis determining whether the median of the 2019-2020 season or the median of the 2016-2017 season is higher. Explain what influenced your hypothesis.

d. Compare your predictions with your partner's predictions. Make any adjustments needed based on your conversation.

2. The table shows the list of the top 15 women goal scorers during the 2016-2017 NCAA Women's Division 1 soccer season.

Rank	Player	Goals
1	Savannah Jordan, Florida	81
2	Tabitha Tindell, FGCU	66
3	Rachel Hill, UConn	61
4	Mckenzie Meehan, Boston College	58
5	Aaliyah Lewis, Alabama St.	57
6	Tyler Lussi, Princeton	53
7	Valerie Sanderson, Memphis	52
8	Murielle Tiernan, Virginia Tech	49
9	Ashley Hatch, BYU	47
10	Ariela Lewis, Alabama St.	45
11	Emily Gingrich, Saint Joseph's	43
12	Margaret Purce, Harvard	42
12	Jaycie Johnson, Nebraska	42
14	Hayley Dowd, Boston College	39
15	Rachel Holden, North Texas	38
15	Lauren Miller, North Dakota St.	38
15	Abby Reed, DePaul	38

a. Find the mean number of goals for the 2016-2017 soccer season.

b. Interpret the mean.

c. Find the median number of goals for the 2016-2017 soccer season.

d. Interpret the median.

e. Complete the statements.

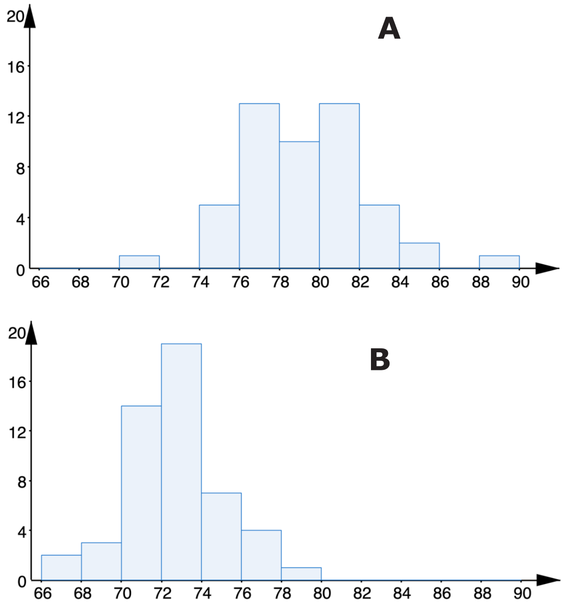
The mean of the 2016-2017 soccer season is
○ less
○ greater
than the mean of the 2019-2020 soccer season. The value in the
○ 2016-2017
○ 2019-2020
soccer season that likely influenced the mean is _________.

The median of the 2016-2017 soccer season is
○ less
○ greater
than the median of the 2019-2020 soccer season.

9.8.5 Wrap-Up: Ticket Out the Door

1. Professional basketball players tend to be taller than professional baseball players.

   The two histograms, labeled A and B, show height distributions of 50 male professional baseball players and 50 male professional basketball players.



Height (in inches)

   a. Choose one of the titles: "Heights of Baseball Players" or "Heights of Basketball Players" for each histogram.

Histogram	A	B
Best Title		

   b. Describe the spread and shape of the distribution of the heights of the basketball players.

Unit 9, Lesson 9: Creating Box Plots

Benchmarks

MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data.

MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts**.**

Learning Targets

- ☐ I can find key values given a box plot within a real-world context.
- ☐ I can create a box plot to represent data.